

Execution-order comparison

Pipeline: RAG | 2 runs | 4 identical question texts | Reference: R1

Reference selected automatically: its actual execution sequence matches the selected questions in questionbank.xlsx sheet/row order.

Run	Folder / model
R1	exp2_rag_2026-10-10_12-19-41 granite4:3b-h
R2	exp2_rag_2026-10-10_13-43-07 granite4:3b-h

Main observations

These are observed differences, not a causal estimate of the effect of shuffling. Output length, background activity and measurement variation may contribute.

Recorded model names match; this does not establish identical model versions or generation settings.

Generation settings, token counts and RAG index identity are not consistently recorded by these reports; equality of those conditions cannot be verified.

Lowest observed emissions: R2, 1.3533 g CO₂e. Spread between highest and lowest: 0.050816 g (3.755% relative to the lowest).

R2 versus R1: different execution sequence. 4 valid successful question pairs; 3 lower, 1 higher, 0 unchanged emissions.

Largest absolute change: Q2, -0.030293 g; execution position 2 to 2. Net change over these matched successful pairs: -0.050816 g.

R2: energy_wh change summed over 4 matched successful pairs: -0.071227.

R2: latency_s change summed over 4 matched successful pairs: 36.612.

Recorded metadata differences: Ordering, Seed, Runtime (s). Unknown or unrecorded values do not establish equivalence.

Historical zero defaults cannot be distinguished from measured zeros. Missing measurements are excluded and partial sums labelled. Percentage changes with a zero reference are N/A.

Recorded shuffling schemes

These rules and seeds come from saved ordering metadata, with CSV metadata used when available. Missing rules are not inferred from the observed sequence. A shuffled scheme can sometimes produce the same sequence.

Run	Recorded scheme	Seed
R1 (reference)	Keep original order	Not applicable
R2	Freely shuffle all selected questions regardless of question type	288198179

Execution sequence comparison

Positions 1-4. Reference: R1. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Position	R1 - Reference Keep original order Seed: Not applicable	R2 Freely shuffle all selected questions regardless of question type Seed: 288198179
1	= Q1 Reference	↑ Q4 Earlier by 3
2	= Q2 Reference	= Q2 Same position
3	= Q3 Reference	= Q3 Same position
4	= Q4 Reference	↓ Q1 Later by 3

Run cost and coverage

Observed sums include failed requests. Coverage is valid readings / all questions. Partial sums cannot be treated as complete batch costs.

Run	Emissions (g)	Energy (Wh)	Request time (s)	Failures / unknown
R1	1.4041 [4/4]	1.9681 [4/4]	232.76 [4/4]	0
R2	1.3533 [4/4]	1.8968 [4/4]	269.37 [4/4]	0

Batch costs: R1-R2

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R1		1.404
	Valid 4/4	
R2		1.353
	Valid 4/4	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R1		1.968
	Valid 4/4	
R2		1.897
	Valid 4/4	

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R1	R2
Folder *	exp2_rag_2026-10-10_12-19-41	exp2_rag_2026-10-10_13-43-07
Model(s)	granite4:3b-h	granite4:3b-h
Pipeline	RAG	RAG
Ordering *	Keep original order	Freely shuffle all selected questions regardless of question type
Seed *	Not recorded	288198179
Questions	4	4
Runtime (s) *	247.48911952972412	284.82679080963135
Completion evidence	Summary present	Summary present
codecarbon_version	"3.3.0"	"3.3.0"
os	"Windows-11-10.0.26200-SP0"	"Windows-11-10.0.26200-SP0"
effective_scope	"machine"	"machine"
measure_power_secs	1	1
pue	1.0	1.0
country_iso_code	"IND"	"IND"
config_profile	"inference"	"inference"
configured_settings	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}
hardware	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}
backends	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}
hardware_evidence	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]

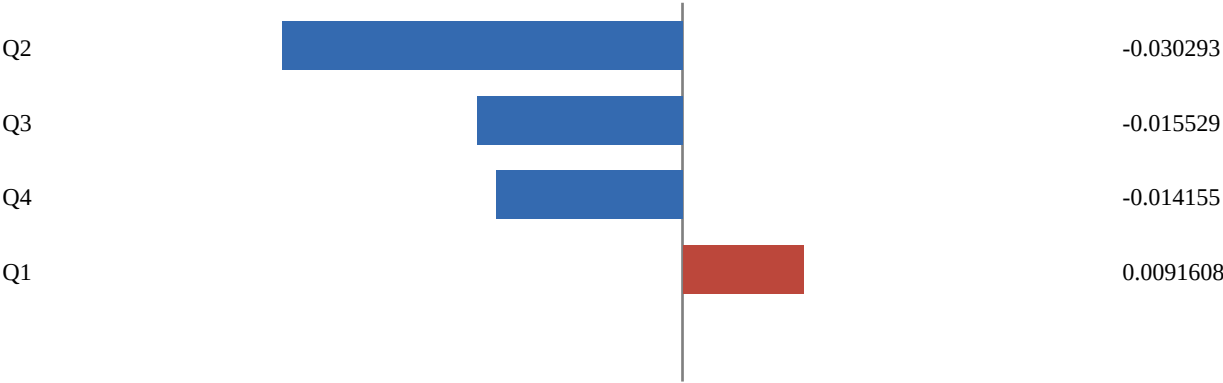
Question emissions change (%)

Relative to R1. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2
Q1	0	2.2791
Q2	0	-30.792
Q3	0	-3.4186
Q4	0	-3.1489

Largest changes: R2 versus R1

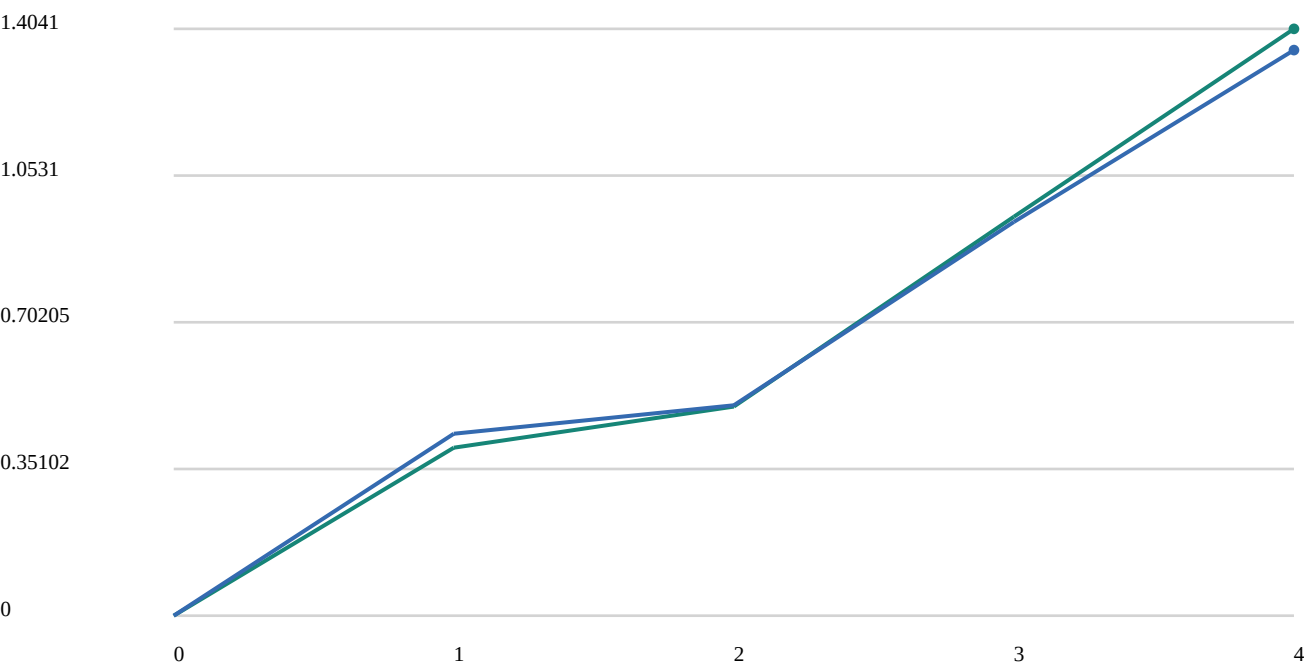
Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.



Cumulative emissions through execution

X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.

R1: 4/4 questions accumulated
R2: 4/4 questions accumulated



Aligned question comparison

Q identifiers follow the reference run. Text matching is exact. Mean emissions are calculated per question across all selected runs, using recorded values only; missing readings are excluded and the contributor count is shown. Recorded costs for failed attempts are included. Percentage changes are unavailable for a zero reference.

Q1: Imagine a customer writes an angry complaint claiming their newly purchased software license key is showing as 'Invalid.' What is the first diagnostic step?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	1	ok	0.40195	0.40653 (2/2)	0.5634	66.227	0	0
R2	4	ok	0.41111	0.40653 (2/2)	0.57624	82.442	0.0091608	2.2791

Q2: If the system shows the key was already activated from a different IP address, how do you communicate this without accusing the customer of fraud?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	2	ok	0.098378	0.083232 (2/2)	0.13789	15.786	0	0
R2	2	ok	0.068085	0.083232 (2/2)	0.09543 2	12.908	-0.030293	-30.792

Q3: The customer responds furiously, demanding an immediate refund. What de-escalation techniques should be used here?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	3	ok	0.45426	0.44649 (2/2)	0.63671	75.83	0	0
R2	3	ok	0.43873	0.44649 (2/2)	0.61495	87.918	-0.015529	-3.4186

Q4: What backend policy checks must be passed before a replacement key or refund can be securely issued?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	4	ok	0.44951	0.44243 (2/2)	0.63006	74.913	0	0
R2	1	ok	0.43535	0.44243 (2/2)	0.61022	86.1	-0.014155	-3.1489

Per-question RAG phase emissions

For each identical question, compare retrieval across runs and generation across runs. All values are recorded emissions in g CO2e, not reference-run differences. N/A means missing; zero remains zero. * marks recorded costs for failed or unknown requests. Cancelled/incomplete batches remain excluded by the existing comparison rules.

Q1: R1-R2

Phase (g CO2e)	R1 Position 1 ok	R2 Position 4 ok
Retrieval	0.0013386	0.00045229
Generation	0.40061	0.41066

Q2: R1-R2

Phase (g CO2e)	R1 Position 2 ok	R2 Position 2 ok
Retrieval	0.00041906	0.0003786
Generation	0.097959	0.067707

Q3: R1-R2

Phase (g CO2e)	R1 Position 3 ok	R2 Position 3 ok
Retrieval	6.5153e-06	8.1639e-06
Generation	0.45425	0.43872

Q4: R1-R2

Phase (g CO2e)	R1 Position 4 ok	R2 Position 1 ok
Retrieval	4.4134e-06	0.0011635
Generation	0.4495	0.43419

RAG phase emissions overview

Q1-Q4; R1-R2. Darker cells mean higher absolute emissions. Retrieval and generation use the same scale across every panel and page. Cell labels use three significant digits; tables above use five. * includes failed/unknown attempt costs. N/A is not zero.

Retrieval (g CO2e)			Generation (g CO2e)		
	R1	R2		R1	R2
Q1	0.00134	0.000452	Q1	0.401	0.411
Q2	0.000419	0.000379	Q2	0.098	0.0677
Q3	6.52e-06	8.16e-06	Q3	0.454	0.439
Q4	4.41e-06	0.00116	Q4	0.45	0.434



Question label key

Q labels follow the execution order in R1. The same exact question text has the same label in every run and throughout this PDF.

Label	Exact question text
Q1	Imagine a customer writes an angry complaint claiming their newly purchased software license key is showing as 'Invalid.' What is the first diagnostic step?
Q2	If the system shows the key was already activated from a different IP address, how do you communicate this without accusing the customer of fraud?
Q3	The customer responds furiously, demanding an immediate refund. What de-escalation techniques should be used here?
Q4	What backend policy checks must be passed before a replacement key or refund can be securely issued?